

MIDTERM PROJECT

ECO403/503–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

FALL 2025

Midterm Project Rules

Please, read this carefully before attempting anything! The midterm project consists of a single analytical problem to be completed in groups and presented as a written report.

Collaboration and Submission

- Students must work in **groups of up to 5 members**. A deviation of +1 is allowed *only with prior written consent* from me.
- Only **one written report** is required per group.
- Each group must send **one email** to me stating the **full list of group members**.
- **Inter-group consultation is permitted**, but **each group must write their own solutions independently**.
- Students are **strictly prohibited** from discussing this project with **anyone outside the class**.
- While there is no oral examination, **students may be called individually to my office** if there is any concern regarding excessive use of AI tools or unclear authorship.
- **All written submissions must be typed**, including all mathematical expressions.
 - It is strongly recommended to use $\text{\LaTeX}$ (e.g., Overleaf) or Microsoft Word with an equation editor.
 - **Handwritten mathematics embedded as images will not be accepted.**
 - This is a formal report and should be treated accordingly.
 - If required, the submission must consist of a single PDF file sent to my email.
 - Your submission must be received **before 23:59 on November 9, 2025, through Moodle. Only one submission per team.**
 - Your written solutions should be **careful, rigorous, and succinct**. There are **penalties for unnecessarily long or poorly structured arguments**. Knowing how to express a mathematical argument clearly and concisely is part of what is being evaluated.

Use of Resources

- You may consult:
 - Your own class notes and printouts of the slides.
 - Any book listed as a reference in the syllabus.
 - Any other book or online source you deem appropriate.
- **All references must be properly cited and acknowledged.**
- You may use **any computer algebra system** (e.g., Mathematica, WolframAlpha, Maple) for final calculations. However, you must provide a **complete logical argument**, omitting only routine algebra when appropriate.

Use of Generative AI Tools

- Use of **Generative AI tools** (e.g., ChatGPT, Claude, Copilot, DeepSeek, etc.) is permitted **only up to 25%** of your written submission.
- This limit applies to both text and mathematical derivations.
- You are fully responsible for the **correctness and understanding** of any AI-generated content.
- Students may be called individually to verify their understanding of any part of the submission. If it becomes clear that AI did the thinking, **this constitutes academic dishonesty**.

Integrity and Consequences

- **Uploading the project paper to any internet repository or database is strictly prohibited.**
- **Any violation** of these rules will be penalized under the Academic Integrity Policy. Even a **Category A violation** may result in **failing the course**, at my discretion and that of the Academic Discipline Committee.
- Random authorship checks may be conducted to ensure compliance.

1 How to Lose Your Degree Because of Cheating: A Game Theoretic Analysis of NU's Academic Disciplinary Committee

A student in ECON403/503 has been suspected of academic dishonesty. Dr. A, bound by both moral duty and academic protocol to make life difficult for such students, reports the case to the appropriate academic disciplinary committee. According to Nazarbayev University's academic integrity policy, the case is then referred to a jury consisting of two faculty members from the School of Sciences and Humanities (SSH), who must evaluate the situation and reach a verdict. The committee requires a *unanimous verdict*: the student is convicted only if both jurors vote to convict. Otherwise, the student is acquitted.

Each juror receives a private signal about the case. This signal is intended to reflect underlying evidence of the student's guilt or innocence. Let the state of nature be denoted $\theta \in \{G, I\}$, where G indicates that the student is *guilty* and I indicates that the student is *innocent*. The prior probability of guilt is denoted $\mu = \Pr(\theta = G)$, where $\mu \in (0, 1)$.

Each juror privately observes a signal $s_i \in \{A, E\}$, where A denotes an *accusatory* signal and E denotes an *exculpatory* signal. The joint distribution of signals (s_1, s_2) depends on the true state θ , and is parameterized by three quantities. The parameter $p \in (1/2, 1)$ captures the likelihood of receiving an accusatory signal when the student is guilty, while $q \in (1/2, 1)$ captures the likelihood of receiving an exculpatory signal when the student is innocent. The parameter $\xi \in [\underline{\xi}, \bar{\xi}]$ governs the degree of correlation between the jurors' signals, with the lower and upper bounds defined by the requirement that all joint probabilities lie in the unit interval.

The joint distribution of signals conditional on the state of nature is given below.

$\theta = G$	$s_2 = A$	$s_2 = E$	$\theta = I$	$s_2 = A$	$s_2 = E$
$s_1 = A$	$p^2 + \xi$	$p(1-p) - \xi$	$s_1 = A$	$(1-q)^2 + \xi$	$(1-q)q - \xi$
$s_1 = E$	$p(1-p) - \xi$	$(1-p)^2 + \xi$	$s_1 = E$	$(1-q)q - \xi$	$q^2 + \xi$

After receiving their respective signals, each juror must take an action $a_i \in \{0, 1\}$, where action 1 denotes a vote to *convict* and action 0 a vote to *acquit*. A conviction occurs if and only if both jurors vote to convict.

Jurors receive payoffs based on the true state and the final verdict. If the student is guilty and convicted, or innocent and acquitted, the juror receives a payoff of 0. If the student is *innocent and convicted*, the juror incurs a cost of $-z$. If the student is *guilty and acquitted*, the juror incurs a cost of $-(1-z)$, where $z \in (0, 1)$.

- 1.1. Define a state of the world to be the triple $\omega = (\theta, s_1, s_2)$, where $\theta \in \{G, I\}$ is the true state of nature, and $s_1, s_2 \in \{A, E\}$ are the private signals received by the two jurors. Let Ω denote the set of all such states. Provide a complete description of the belief space by explicitly writing the prior distribution over Ω , and defining the signalling functions that map each state $\omega \in \Omega$ to the signal observed by each juror.
- 1.2. Complete the description of the Bayesian game by writing down the payoff function of each juror as a function of the state $\omega = (\theta, s_1, s_2)$ and the action profile. Describe formally the set of pure strategies of each player.
- 1.3. Let r denote the posterior probability that the student is guilty, given the juror's information. Compute the expected payoff to a juror when the defendant is acquitted, and the expected payoff when the defendant is convicted. Compare these expressions and interpret the role of the parameter z in determining the juror's preferences over the two outcomes.
- 1.4. Consider the conditional distributions of signals given the state $\theta \in \{G, I\}$. Compute the values of $\underline{\xi}$ and $\bar{\xi}$ that ensure the resulting joint distributions are valid probability distributions; that is, all entries lie in the interval $[0, 1]$.
- 1.5. For the remainder of the problem, unless otherwise specified, assume $\xi = 0$; that is, the signals are conditionally independent given the state. Assume further that the signals are ordered as $A > E$.

Consider the symmetric strategy profile in which each juror follows a monotonically increasing strategy based on their signal: voting to acquit when receiving signal E , and voting to convict when receiving signal A . Derive the conditions on the parameter z under which this strategy profile constitutes a Bayesian Nash Equilibrium in pure strategies.

- 1.6. Recall that a *type I error* occurs when an innocent defendant is convicted, while a *type II error* occurs when a guilty defendant is acquitted. Under the symmetric equilibrium described in the previous item, where each juror votes to acquit upon receiving signal E and to convict upon receiving signal A , compute the joint probability mass function over the state of nature and the equilibrium action profile; that is, compute the probability of each tuple (ω, a_1^*, a_2^*) , where a_i^* denotes the equilibrium strategy of juror i . Using this joint pmf, compute the probability that the committee makes a type I error and the probability that it makes a type II error.
- 1.7. Now suppose that $\xi \in [\underline{\xi}, \bar{\xi}] \setminus \{0\}$; that is, the signals are no longer conditionally independent given the state. Repeat the previous two items: (i) derive the conditions on z under which the symmetric monotone strategy profile remains a Bayesian Nash Equilibrium, and (ii) compute the probability that the committee makes a type I or type II error under this equilibrium. Compare these results to those obtained under $\xi = 0$. What role does the correlation parameter ξ play in shaping equilibrium incentives and institutional accuracy?